2. A group of students investigate how the surface area of a falling paper cake case affects its terminal speed.

- Cake case 1 has a mass of 0.5 g and a surface area of 100 cm^2.
- Cake case 1 is dropped from a height of 1.80 m but only timed over the final 1.50 m of the fall.

The students' results are shown in the table below.

Drop time (s)			Mean drop time (s)	Drop distance (m)
Attempt 1	Attempt 2	Attempt 3		
0.96	0.92	0.94	....	1.50

(a) (i) The students decide there are no anomalies. Explain why. [1]

....

(ii) **Complete the table** to show the mean drop time. [1]
Space for calculation.

(b) The experiment is repeated with cake case 2.
It has the same shape and the same mass as cake case 1.
However, cake case 2 has a surface area of 50 cm^2.
The students correctly calculate the terminal speed for both cake cases.

Cake case 1		
Mass (g)	Surface area (cm^2)	Terminal speed (m/s)
0.5	100	1.6

Cake case 2		
Mass (g)	Surface area (cm^2)	Terminal speed (m/s)
0.5	50	2.3

Examiner only

(i) A cake case reaches terminal speed when its weight is balanced by air resistance. **Tick (✓)** the **three** correct statements. [3]

Cake case 2 has the same terminal speed as cake case 1. ☐

Cake cases 1 and 2 have identical weight. ☐

At terminal speed, cake case 1 experiences a greater value of air resistance than cake case 2. ☐

At terminal speed, both cake cases experience identical values of air resistance. ☐

At terminal speed, cake case 1 experiences a smaller value of air resistance than cake case 2. ☐

At terminal speed, both cake cases have zero acceleration. ☐

(ii) Before the experiment was carried out the students made the following prediction:

"If the surface area of the cake case is halved its terminal speed will double."

Use data from the tables on the previous page to explain whether their prediction was correct. [2]

....

....

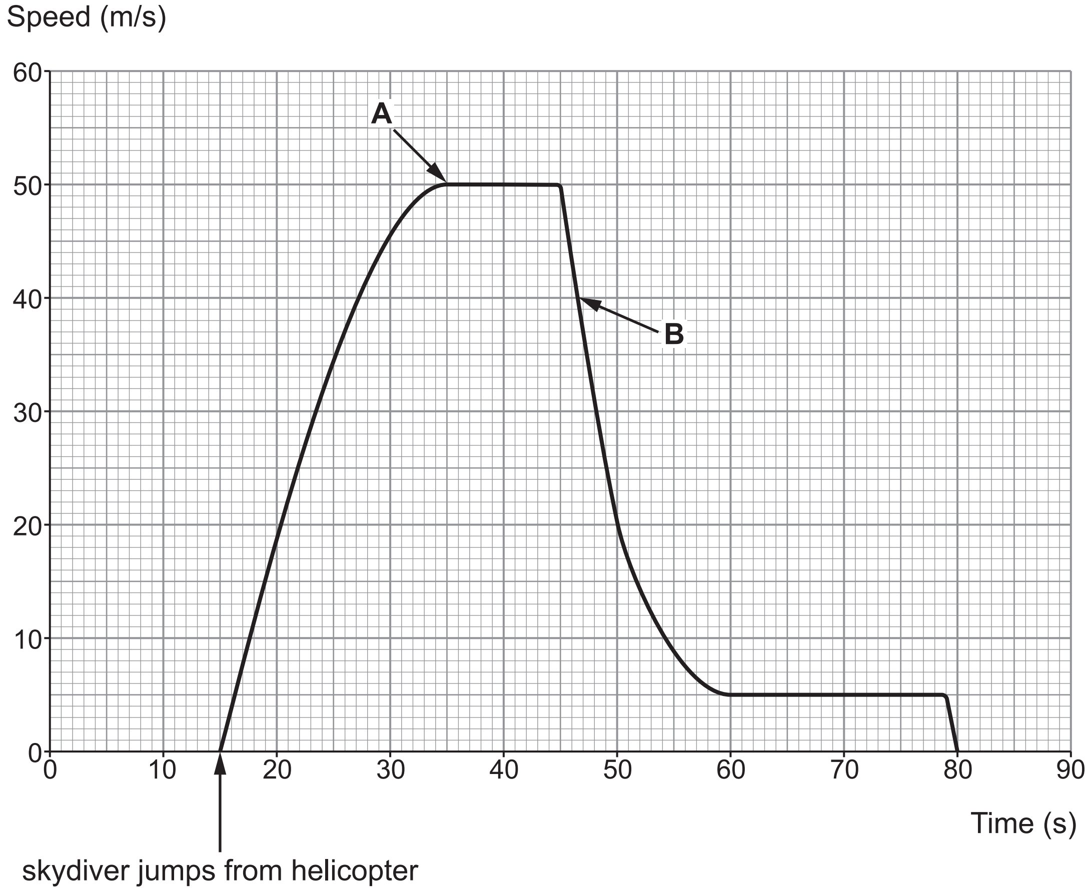
....

....

3420UD01
07

Examiner only

(c) A skydiver sits at the doorway of a helicopter for 15 s before jumping. His speed is recorded and displayed on the graph below.



Tick (✓) the **three** correct statements. [3]

- The skydiver lands on the ground 80s after jumping from the helicopter. ☐
- The terminal speed after the parachute is opened is $\frac{1}{10}$th of the terminal speed before the parachute is opened. ☐
- The skydiver's weight is greatest at the point labelled **B** on the graph. ☐
- The parachute is opened 30 s after the skydiver leaves the helicopter. ☐
- At point **B** the weight of the skydiver is greater than the air resistance. ☐
- At point **A** the skydiver stops accelerating. ☐

10